Communication and Information Acquisition in Committees

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Introduction

Introduction

- Many collective decisions rely on *costly* information acquisition by individual members.

  Examples: Jurors review evidence; hiring committee members evaluate candidates' profiles

- Information is a public good. $\rightarrow$ Members may free ride, and information is underprovided.

  (e.g., Persico 2004; Gerardi and Yariv 2008; Bhattacharya et al. 2017; Elbittar et al. 2020)

- Pre-vote communication is common in these settings.

  Examples: Jurors deliberate; committee members discuss in meetings

- RQ: How does communication affect this free-riding problem?
  1. Does it increase or decrease information acquisition? *(Key question)*
  2. How does it affect group accuracy and welfare?

Preview of this paper

Setting: a common-interest group decides by majority voting; info is costly

Institution comparison:

$$\text{Information acquisition} \longrightarrow \text{Voting} \quad \text{(e.g. Persico 2004)}$$

vs.

$$\text{Information acquisition} \longrightarrow \text{Cheap talk communication} \longrightarrow \text{Voting.}$$

Main findings:

- **Theory**: Communication can increase or decrease information acquisition, depending on information costs. Likewise for group accuracy and welfare.
- **Experiment**: Communication increases acquisition and group outcomes. The downside is missing as subjects overvalue their own acquisition under communication.

Contributions

The first paper to show **how pre-vote communication affects information acquisition**.

1. Information acquisition in collective decisions
   - Prior work: group size and voting rules. e.g. Persico 2004; Gerardi and Yariv 2008; Martinelli 2007; Elbittar et al. 2020; Meyer and Rentschler 2023
   - This paper: pre-vote communication is another design margin

2. Deliberation in collective decisions
   - Prior work: aggregation of given information. e.g. Coughlan 2000; Gerardi and Yariv 2007; Goeree and Yariv 2011; Iaryczower et al. 2018
   - This paper: deliberation also affects costly information provision!

Theory

Setup

- A three-player group decides between options A and B via majority voting.
  ($N = 2n + 1$ in the paper.)
- State A or B occurs equally likely. Players do not observe the state but have correct prior.
- Players have common interests: everyone receives payoff 1 if the group decision matches the true state, and 0 otherwise.

Setup: Two games

C: Information acquisition ⟶ Communication ⟶ Voting

NC: Information acquisition ⟶ Voting

- **Information acquisition:** Each player can pay a cost $c > 0$ for an *i.i.d.* binary signal of accuracy $q \in (0.5, 1)$. Acquisition decisions and signals are private.
- **Communication:** cheap talk message $m \in M$, with $|M| \geq 3$
- **Voting:** simultaneous voting for A or B, no abstention; majority rule.

Equilibrium concept and post-acquisition subgame

I focus on *symmetric, neutral* PBEs under two restrictions:

- weakly undominated voting strategies (as in the voting literature)
- maximal information transmission (as in the cheap-talk literature)

In equilibrium,

- **Sincere voting:** vote according to the posterior (randomize between two votes equally when uninformed in NC)
- **Full revelation:** communication makes all private information common knowledge

Equilibria are characterised by information acquisition strategies.

Core Intuition: Why communication matters?

Given the others' strategies, a player's incentive to buy signal:

$$U(Buy) - U(NotBuy) = \underbrace{Pr(piv) \times (q - \frac{1}{2})}_{\text{Expected increase in group accuracy}} - c$$

$Pr(piv)$: probability that her signal is pivotal

- In NC: if the others' votes are balanced, i.e., if $v_1 \neq v_2$
- In C: if the others' signals are balanced, i.e., if $s_1 \neq s_2$ or $s_1 = s_2 = \emptyset$

$\rightarrow$ not align in the two games!

Communication's effect on incentive

Case I: Neither P1 nor P2 buys.

- In C, P3's signal determines the group decision.
- In NC, P3's signal matters only if $v_1 \neq v_2$.

Persuasion effect (+): Communication amplifies **one's own signal's** influence, making acquisition more valuable.

- In NC, a signal affects only one's own vote. Communication allows it to sway others' votes.

Communication's effect on incentive

Case II: One buys, one does not. E.g., P1 buys, P2 does not, $q \approx 1$ ▸ same for general q

- In C, P3's signal adds no value.
- In NC, P3's signal matters if $v_1 \neq v_2$.

Substitution effect (-): Communication amplifies **others' signals'** influence, making one's own signal less necessary.

- In NC, uninformed votes may offset informed votes. Communication removes this noise.

Communication's effect on incentive

Case III: Both P1 and P2 buy.

- In C, P3's signal matters if $s_1 \neq s_2$.
- In NC, P3's signal matters if $v_1 \neq v_2$. This is equivalent to $s_1 \neq s_2$.

$\rightarrow$ No effect of communication in this case.

Summary:

1. Neither buys: Persuasion effect (+)
2. One buys, one does not: Substitution effect (-)
3. Both buy: no effect

Value of acquisition

Given each other voter acquires information with probability $\alpha \in [0, 1]$:

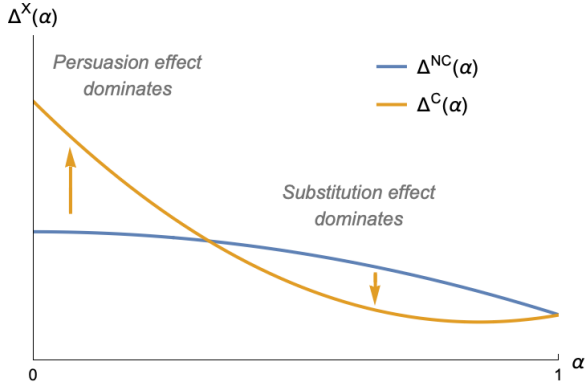
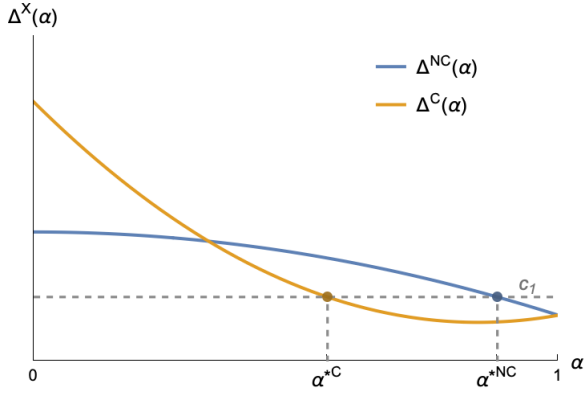


Figure 1: Value of information acquisition

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Value of acquisition and equilibrium

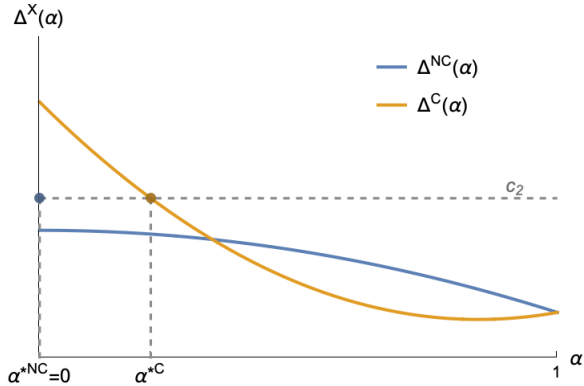
- Equilibra when $c = c_1$



$\rightarrow \alpha^{*C} < \alpha^{*NC}$

Value of acquisition and equilibrium

- Equilibria when $c = c_2$



$\rightarrow \alpha^{*C} > \alpha^{*NC}$

Equilibrium results

Communication's effect on acquisition depends on c:

- For relatively low c, communication reduces info acquisition.
- For relatively high c, communication increases info acquisition.

	Low cost	High cost
NC	high acquisition	low acquisition
Effects of C	persuasion ↑, **substitution** ↓↓	**persuasion** ↑↑, substitution ↓

Effect on group accuracy and welfare:

- Same two-sided pattern, with different c cutoffs. ▸ More ▸ Figure

Experimental Design

Treatments and main hypotheses

4 treatments: $\{\text{NC, C}\} \times \{\text{Low } c, \text{ High } c\}$; $N = 3$, $q = 0.9$

	Low cost ($c = 0.10$)	High cost ($c = 0.25$)
NC	0.884	0
C	0.560	0.215

Table 1: Equilibrium information acquisition rates

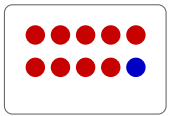
Main hypotheses:

H1 At low info cost, communication reduces info acquisition

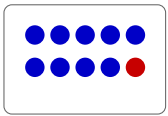
H2 At high info cost, communication increases info acquisition

Experimental Task





Red jar



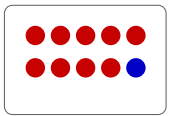
Blue jar

Jar selected

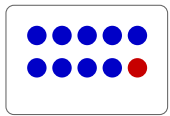
- The computer secretly selects one jar at random (50–50).

Experimental Task





Red jar



Blue jar

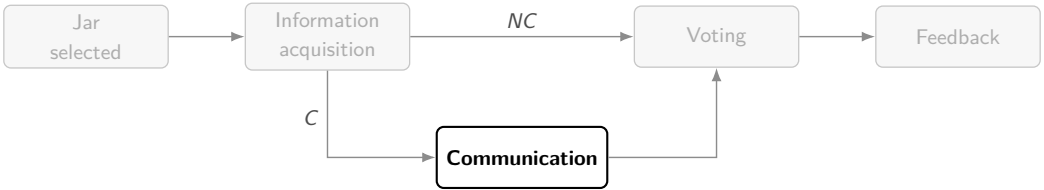
Jar selected

- The computer secretly selects one jar at random (50–50).

Information acquisition

- Each member decides whether to pay a cost c to draw a ball ($q = 0.9$).

Experimental Task



Communication

"I have drawn a red ball"

"I have drawn a blue ball"

No message

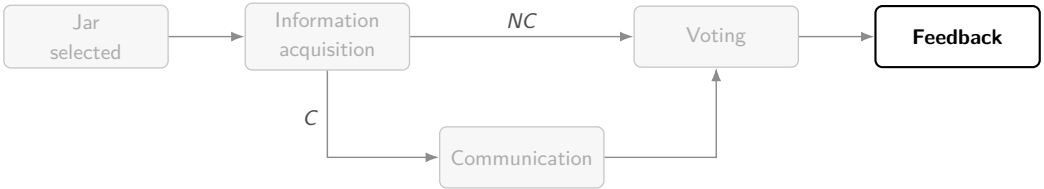
Experimental Task



Voting

- Each member votes for Red or Blue.
- Group decision is determined by majority rule.

Experimental Task



Voting

- Each member votes for Red or Blue.
- Group decision is determined by majority rule.

Feedback

- Participants see the true jar, group decision, vote counts, and the round payoff.

Procedures and Measures

Procedures

- Between-subject design.
- 30 rounds; random rematching within fixed matching groups of six.
- One randomly selected round paid.
- 288 participants; Tilburg CentERlab, 2025; pre-registered.

Procedures and Measures

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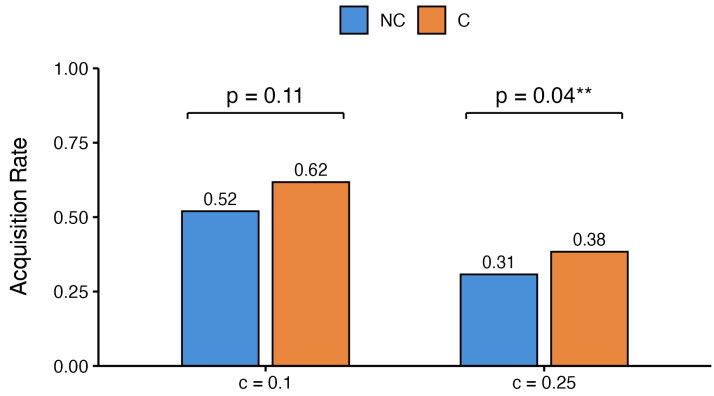
Additional measures

- perceived value of acquisition, beliefs about others' acquisition, and conditional choices.
- risk preference, uncertainty attitude, social preference, demographics

Experimental Results

Communication boosts information acquisition

- Communication increases acquisition when $c = 0.25$ (support H2).
- No significant difference when $c = 0.1$ (if anything, opposite to H1).



MW test; one-sided for $c = 0.25$, two-sided for $c = 0.1$.

▸ Individual ▸ By-round ▸ Regression ▸ Message and vote

Communication enhances group outcomes

- Communication improves group accuracy and welfare at both costs.

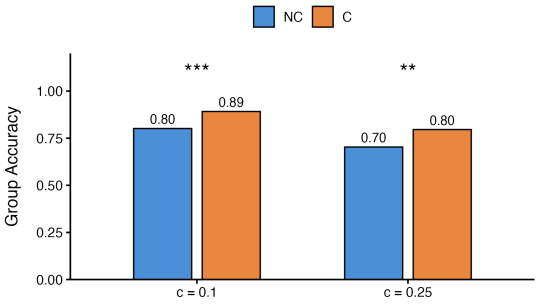


Figure 2: Rate of correct group decisions

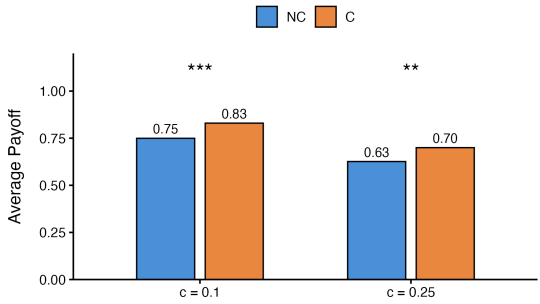
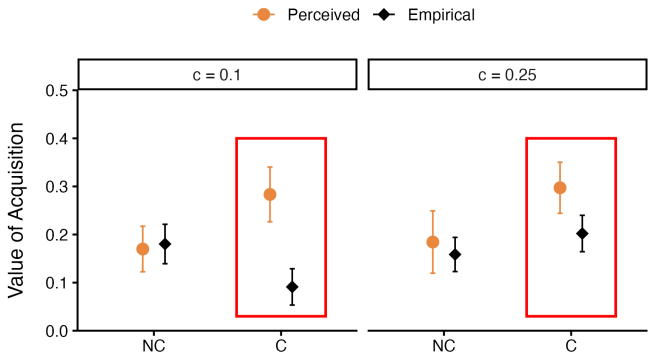


Figure 3: Payoff

Why is the Downside Missing?

Perceived value of acquisition (PVA)

- PVA $\equiv$ Pr(group correct | I draw) $-$ Pr(group correct | I don't draw), elicited in round 30.



- Communication increases PVA at both costs

  PVA predicts acquisition.

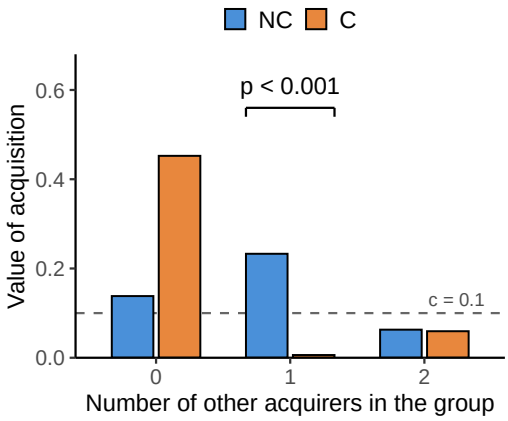
- Subjects overvalue acquisition under communication
  - Why? Not from beliefs about others' acquisition

    Beliefs are broadly accurate.

Subjects respond to persuasion, but not to substitution

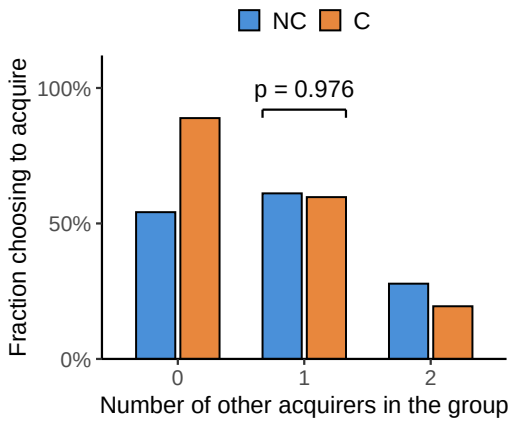
(a) Empirical conditional value

computed from observed group outcomes



(b) Reported acquisition choices

"If 0/1/ 2 others draw, would you draw?" (post-exp)



Conclusion

Conclusion

RQ: How does pre-vote communication affects information acquisition?

Theory: two-sided effects of communication

- Persuasion effect $(+)$ vs. substitution effect $(-)$; which dominates depends on info costs.
- When info is relatively cheap, communication can make everyone worse off.

Experiment: confirms the upside

- Communication increases info acquisition, group accuracy, and welfare.
- Subjects underappreciate substitution effect, overvaluing acquisition under comm.

Implications

- **Institutional design:** turning on/off communication is a lever for shaping info acquisition.
- **Evaluating deliberation:** should not treat the amount of information as fixed.

Thank you! Suggestions welcome!

Appendix

Appendix: No additional value from a second signal

With one signal, a second signal will **not** improve the likelihood that the group selects the correct option. This holds for all q.

- With one signal, the prob that the posterior prefers the truth is

$$q$$

- With two signals, the prob that the posterior prefers the truth is

$$\begin{aligned} & q^2 + 2q(1-q) \cdot \frac{1}{2} \\ =& q^2 + q(1-q) \\ =& q \end{aligned}$$

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Appendix: Expression N=3

Voter 1 and 2 acquire info with prob α. Consider voter 3's incentives.

In game NC,

$$\Delta^{NC}(\alpha) = \left(q - \tfrac{1}{2}\right)\left((1-\alpha)^2 \cdot \tfrac{1}{2} + 2\alpha(1-\alpha) \cdot \tfrac{1}{2} + \alpha^2 \cdot 2q(1-q)\right).$$

In game C,

$$\Delta^{C}(\alpha) = \left(q - \tfrac{1}{2}\right)\left((1-\alpha)^2 \cdot 1 + 2\alpha(1-\alpha) \cdot 0 + \alpha^2 \cdot 2q(1-q)\right).$$

Hence, the difference

$$\Delta^{C}(\alpha) - \Delta^{NC}(\alpha) = \underbrace{\left(q - \tfrac{1}{2}\right)(1-\alpha)^2 \cdot \tfrac{1}{2}}_{\text{persuasion effect}} + \underbrace{\left(q - \tfrac{1}{2}\right) 2\alpha(1-\alpha) \cdot -\tfrac{1}{2}}_{\text{substitution effect}}$$

Appendix: Different n and q

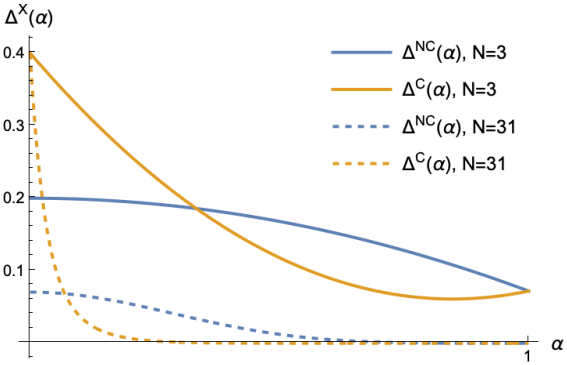


Figure 4: Different n

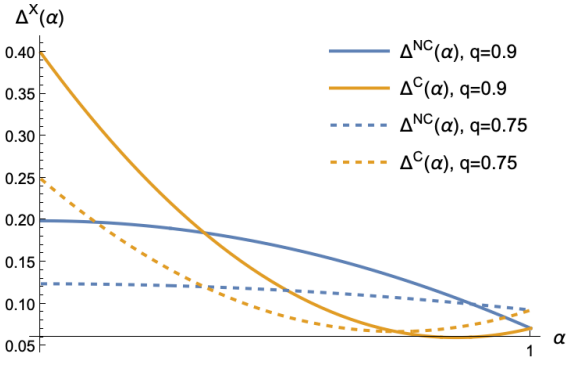
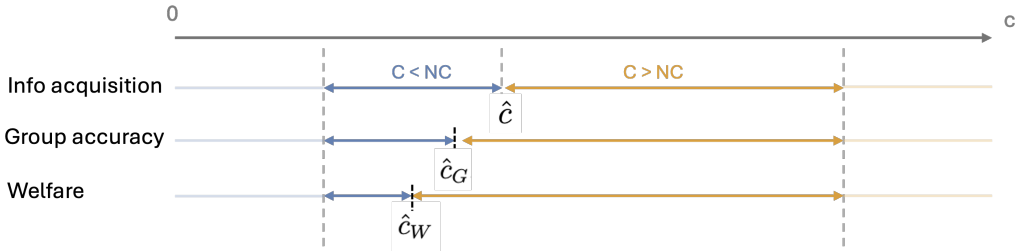


Figure 5: Different q

Appendix: Group accuracy and welfare



- Lower cost cutoffs for G and W than for info acquisition: extra gains come from more efficient information aggregation.
- Cheap-talk communication can be welfare-diminishing.

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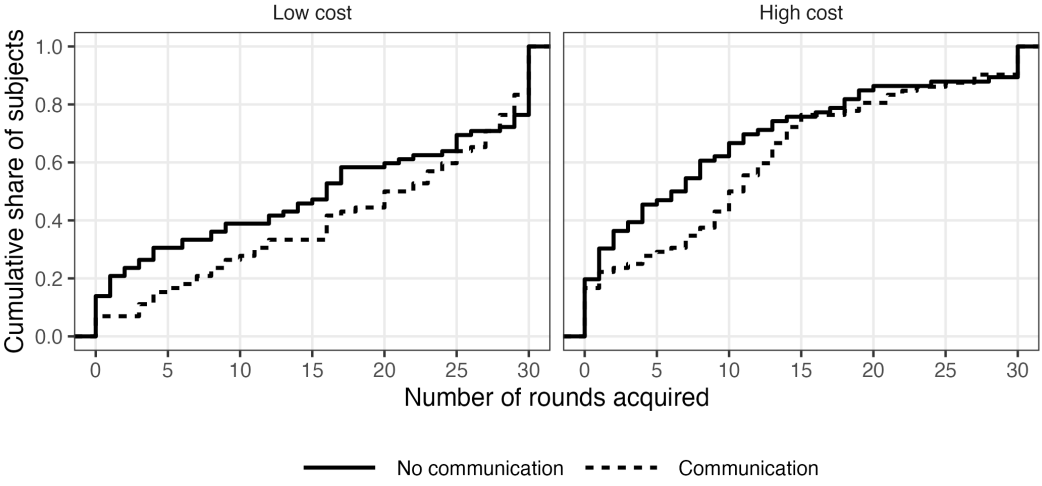
Appendix: More on predictions

	$c = 0.10$			$c = 0.25$		
	m^*	G^*	W^*	m^*	G^*	W^*
NC equilibrium	**0.884**	**0.942**	**0.871**	0	0.500	0.500
C equilibrium	0.560	0.879	0.823	**0.215**	**0.707**	**0.653**

Appendix: Pure-strategy equilibrium

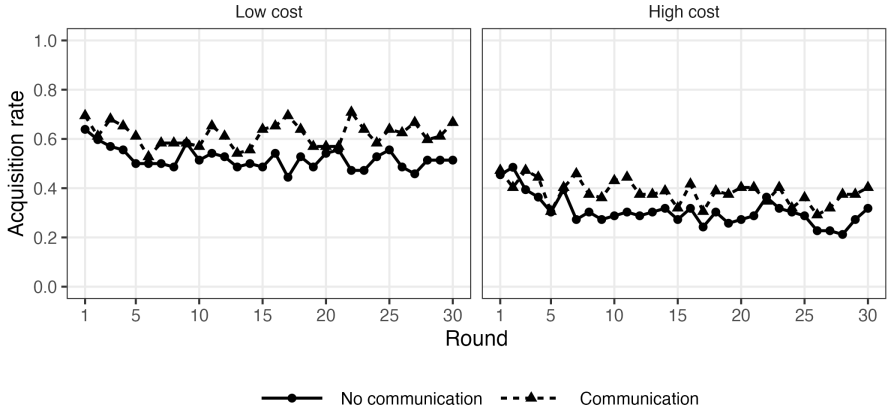
	$c = 0.10$	$c = 0.25$
C	One buys	One buys
NC	Two buy	None buy

Appendix: Information acquisition - individual



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Appendix: Acquisition by round



- $c = 0.1$: First 15 rounds = Last 15 round
- $c = 0.25$: First 15 rounds > Last 15 round, significant Back

Appendix: Regression

Table 2: Effect of Communication on Draw Decisions (OLS)

	Dependent Variable: Draw (0/1)			
	(1)	(2)	(3)	(4)
Comm	0.087**	0.098*	0.114**	0.115**
	(0.035)	(0.052)	(0.054)	(0.048)
High Cost	-0.223***	-0.212***	-0.212***	-0.220***
	(0.034)	(0.055)	(0.055)	(0.051)
Comm × High Cost		-0.021	-0.021	-0.032
		(0.069)	(0.069)	(0.067)
First-half Rounds and × Comm			✓	✓
Individual Controls				✓
Comm effect at High Cost		0.076**	0.092**	0.083**
Observations	8,460	8,460	8,460	8,460

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the matching-group level. Stars for the joint test row based on one-sided tests. Individual controls: uncertainty attitude, social preference, and gender.

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Appendix: Messages and voting

1. Communication is highly informative:
   - draw: no lie
   - not draw: 10% lie at $c = 0.25$, 16% lie at $c = 0.1$

Empirical message accuracy: 86.4% at the low cost and 84.2% at high cost

2. Voting is almost always sincere:
   - In C, when there is a majority message in the group, subjects vote with the majority message in 97% of cases at both c.
   - In NC, subjects vote according to their signal in 99.7% of cases. 50/50 when uninformed.

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Appendix: Regression full results

Effect of Communication on Signal Acquisition

	Dependent variable:				
	Signal Acquire				
	(1)	(2)	(3)	(4)	(5)
Comm	0.087**	0.098*	0.114**	0.115**	0.038
	(0.035)	(0.052)	(0.054)	(0.048)	(0.046)
High cost	-0.223***	-0.212***	-0.212***	-0.220***	-0.231***
	(0.034)	(0.055)	(0.055)	(0.051)	(0.040)
Comm × High cost		-0.021	-0.021	-0.032	-0.032
		(0.069)	(0.069)	(0.067)	(0.065)
Early round			0.039***	0.039***	0.039***
			(0.015)	(0.015)	(0.015)
Comm × Early round			-0.032*	-0.032*	-0.032*
			(0.019)	(0.019)	(0.019)
Perceived value of acquisition					0.669***
					(0.073)
Female				-0.041	-0.026
				(0.045)	(0.038)
Risk tolerance				0.0004	0.001
				(0.001)	(0.001)
Uncertainty tolerance				0.007	0.004
				(0.008)	(0.007)
Pro-sociality				0.006***	0.006***
				(0.001)	(0.001)
Observations	8,460	8,460	8,460	8,460	8,460

Appendix: Regression including PVA

	Dependent Variable: Draw (0/1)				
	(1)	(2)	(3)	(4)	(5)
Comm	0.087**	0.098*	0.114**	0.115**	0.038
	(0.035)	(0.052)	(0.054)	(0.048)	(0.046)
High Cost	-0.223***	-0.212***	-0.212***	-0.220***	-0.231***
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Perceived value of acquisition					0.669***
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First-half Rounds and × Comm			✓	✓	✓
Individual Controls				✓	✓
Comm effect at High Cost		0.076**	0.092**	0.083**	0.006
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Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the matching-group level. Stars for the joint test row are based on one-sided tests. Individual controls: uncertainty attitude, social preference, and gender.

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Appendix: PVA

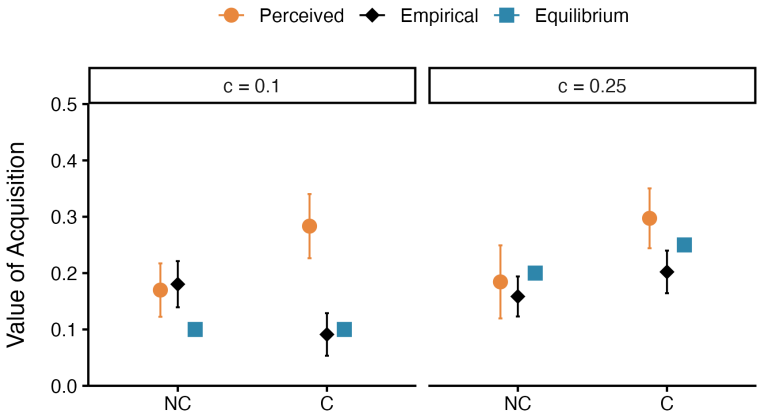
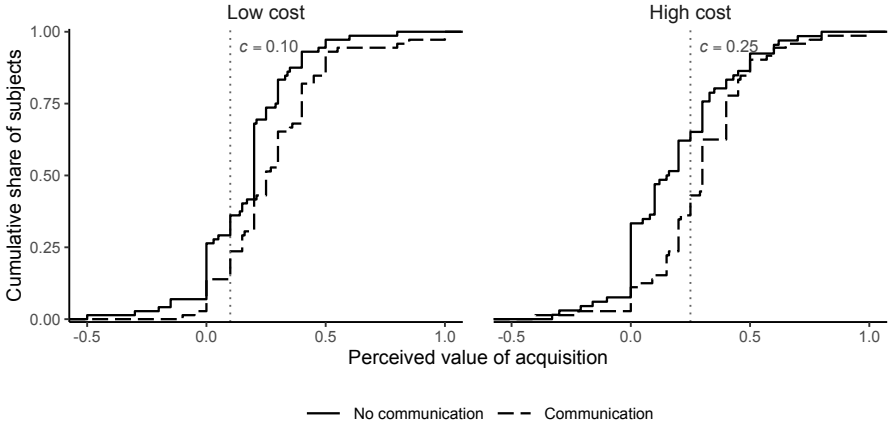


Figure 6: Value of Acquisition: perceived v.s. empirical v.s. equilibrium

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Appendix: distribution of PVA



Appendix: Beliefs about others draw behavior at c=0.1

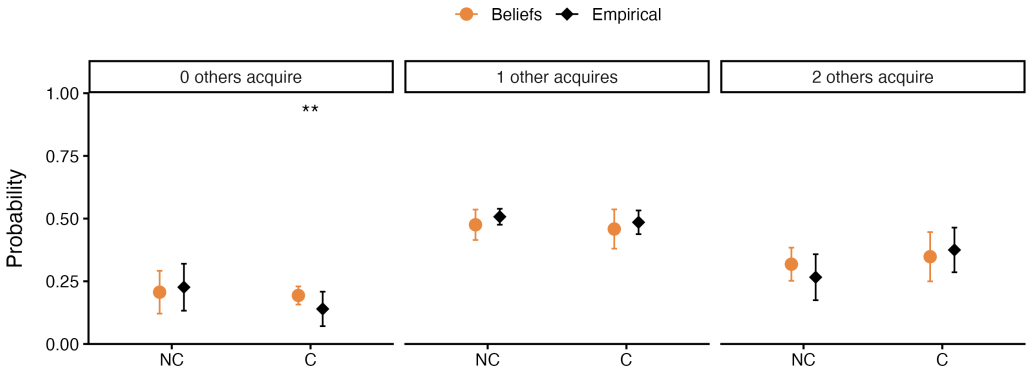


Figure 7: Beliefs about others draw at c=0.1: fairly accurate

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Appendix: Beliefs about others draw behavior at c=0.25

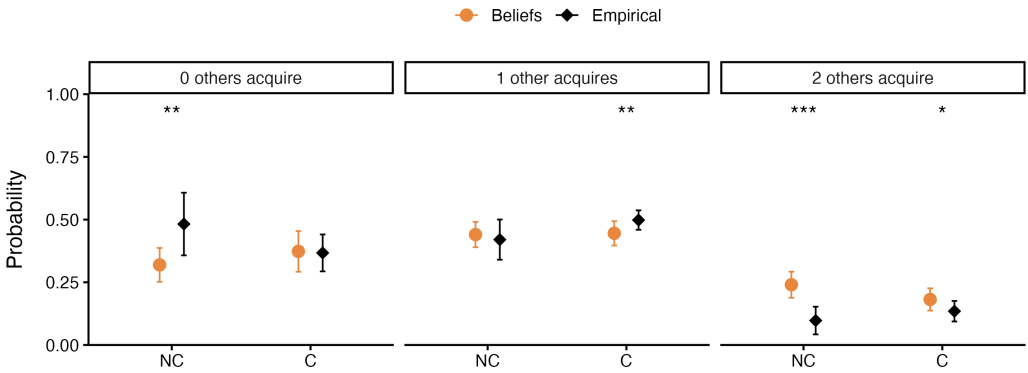


Figure 8: Beliefs about others draw at c=0.25: overestimate

Appendix: Alternative explanations and limitations

- **Equilibrium selection:** Communication is highly informative and voting largely sincere; less-revealing communication or asymmetric acquisition equilibria do not eliminate the two-sided predictions.
- **Risk and social preferences:** Do not explain conditional choices with exactly one other acquirer; risk preferences are not associated with acquisition.
- **Level-k reasoning:** Can eliminate the predicted decrease under certain level-0 assumptions, but does not explain conditional choices with exactly one other acquirer.
- **Quantal response:** Largely explains the more interior acquisition rates, but does not explain the missing downside.

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Appendix: Quantal-response fits

Specification	Estimated parameters	LNC	LC	HNC	HC	RMSE
Observed	–	0.520	0.618	0.308	0.384	–
Nash equilibrium	–	0.884	0.560	0.000	0.215	0.254
Logit QRE	$\hat{\lambda} = 7.556$	0.600	0.521	0.374	0.359	0.072
QRE + CARA	$(\hat{\lambda}, \hat{A}) = (13.620, 5.130)$	0.551	0.533	0.295	0.435	**0.052**
QRE + inequality aversion	$(\hat{\lambda}, \hat{\eta}) = (4.426, 0.999)$	0.580	0.518	0.349	0.343	0.065
QRE + cursed message inference	$(\hat{\lambda}, \hat{\chi}) = (8.970, 0.282)$	0.614	0.599	0.356	0.406	0.055

- **Baseline QRE** captures more interior acquisition rates, but wrong in directions.
- **Extensions:** CARA improves the fit but requires strong risk aversion. Cursed message inference attenuates the low-cost decline and preserves the high-cost increase.